

Throne Ingest PoC One-Pager

What it proves: a small Go/AWS architecture can ingest health-device captures, process them asynchronously, expose operational telemetry, and keep unit economics visible from day one.

Core path: device gRPC upload -> S3 raw bucket -> SQS ingest queue -> processor -> inference service -> Postgres findings + derived S3 artifact -> REST API.

Reliability posture: SLOs cover ingest availability, ingest latency, freshness, and zero acknowledged-data loss. Dashboards surface RED metrics, queue backpressure, DLQ depth, inference p99, and error-budget burn.

Cost posture: the live Grafana cost dashboard converts real bytes/day and captures/day into current-burn \$/user/month. Main levers are S3 lifecycle tiering, compression, ARM/Graviton fleet, Fargate right-sizing, and Spot processor workers.

What is intentionally fake in the PoC: synthetic inference model, no real PHI, single-region dev environment, simplified Grafana pricing constants.